

SUMMARY

Applied mathematician with expertise in Monte Carlo methods, uncertainty quantification, and large-scale numerical simulation for high-dimensional nonlinear partial differential equations with stochastic uncertainty. Experienced in predictive modeling and digital twin frameworks for fusion energy applications. Extensive experience designing statistically efficient estimators that achieve up to two orders-of-magnitude reductions in computational cost while maintaining statistical reliability. Proven ability to build scalable simulation pipelines, develop surrogate and multi-fidelity methods, and translate mathematical theory into robust algorithms.

EDUCATION

University of Maryland, College Park, Maryland Aug 2018 - May 2024

Ph.D. in Applied Mathematics & Scientific Computation, GPA: 3.809/4

Dissertation: Computational Methods for Uncertainty Quantification in Plasma Equilibrium

Courant Institute of Mathematical Sciences, New York University, New York Aug 2016 - May 2018

M.S. in Mathematics, GPA: 3.67/4

Dissertation: Iterative methods for finite element discretization of semi-linear elliptic PDE's: An application to plasma equilibrium

Nanyang Technological University, Singapore Aug 2010 - May 2014

B.S. in Applied Mathematics, GPA: 4.38/5

EXPERIENCE

Rice University Houston, TX

Postdoctoral Researcher (Pfeiffer Postdoctoral Instructor) Aug 2024 – Present

- Developed multi-fidelity model hierarchies for Monte Carlo (MFMC) methods in uncertainty quantification for fusion plasma simulations, achieving 10 ~ 50 times end-to-end runtime reductions relative to direct sampling while preserving estimator accuracy.
- Designed adaptive pilot-sampling algorithms for biased correlation estimation to stabilize MFMC workflows, integrating sensitivity analysis, confidence-interval estimation, and BCa bootstrap techniques.
- Reconstructed the MFMC framework to eliminate degenerate sample allocations, enabling stable and reliable estimation in correlated multi-model regimes.
- Created recursive, integer-constrained MFMC allocation algorithms using dynamic programming, improving budget utilization and estimator accuracy; replaced exhaustive model selection with a backtracking-based strategy, reducing computational complexity.
- Taught graduate and undergraduate level courses in matrix analysis for data science, continuum mechanics, calculus and special topics in uncertainty quantification, emphasizing conceptual clarity, mathematical rigor, and independent problem-solving.
- Communicated complex modeling, numerical methods, and algorithmic results to diverse technical audiences across academia and national laboratories, facilitating effective interdisciplinary collaboration.

University of Maryland, College Park Maryland, MD

Research Assistant Aug 2018 – May 2024

- Developed efficient surrogate models using sparse-grid stochastic collocation and Monte Carlo methods for high-dimensional uncertainty quantification in expensive simulations of the semi-linear free-boundary Grad-Shafranov equation for fusion plasmas, achieving 7 ~ 30 times reductions in computational cost relative to direct simulation.
- Extended multilevel Monte Carlo (MLMC) methods to semi-linear free-boundary PDEs by exploiting variance decay across both uniform and adaptive discretization hierarchies. Demonstrated that uniform refinement yielded 60 ~ 200

times cost reductions but degraded geometric accuracy, while adaptive strategy significantly improved key geometric features at comparable computational cost.

- Designed a hybrid surrogate-based MLMC framework combining sparse-grid surrogates with multilevel sampling, achieving 50 ~ 300 times cost reductions relative to direct Monte Carlo without sacrificing statistical accuracy.
- Collaborated with applied mathematicians and computational scientists on algorithm development, numerical analysis, and large-scale computational studies.
- Engaged in interdisciplinary research activities through graduate research groups, seminars, conferences, and joint projects, strengthening collaborative problem-solving skills and broadening research perspective.

Center for Disruptive Photonic Technologies, Nanyang Technological University

Singapore

Project Officer

Aug 2014 – Jul 2016

- Coordinated cross-functional technical projects involving data reporting, process optimization, and stakeholder communication.
- Managed internal information systems, digital content workflows, and operational logistics supporting research initiatives.

Nielsen

Singapore

Data Science Intern (Final Year Project)

Oct 2013 – May 2014

- Applied genetic algorithm-based optimization methods to improve retail sales sampling strategies.
- Constructed moving-window sampling schemes to improve robustness and stability of trend estimation in consumer time-series data. Translated optimization statistical findings into forecasting insights used for business decision-making.

PUBLICATIONS

- [1] Inverse problem for uncertainty quantification in the Grad-Shafranov free boundary problem. (In preparation)
- [2] Adaptive integer sample sizes estimation for multi-fidelity Monte Carlo, with Matthias Heinkenschloss. (To appear)
- [3] Multi-fidelity Monte Carlo for uncertainty quantification in the Grad-Shafranov free boundary problem, with Matthias Heinkenschloss. (To appear)
- [4] Multi-fidelity Monte Carlo sample allocation revisit and its recursive rounding for sample size estimation. (To appear)
- [5] Surrogate-based multi-level Monte Carlo methods for uncertainty quantification in the Grad-Shafranov free boundary problem, with Howard Elman and Tonatiuh Sánchez-Vizuet. (Accepted by *Computer Physics Communications*, to appear. arXiv: 2501.08482)
- [6] Multilevel Monte Carlo methods for the Grad-Shafranov free boundary problem, with Howard Elman and Tonatiuh Sánchez-Vizuet. (*Computer Physics Communications*, 298 (2024), paper No. 109099.)
- [7] Surrogate approximation of the Grad-Shafranov free boundary problem via stochastic collocation on sparse grids, with Howard Elman and Tonatiuh Sánchez-Vizuet. (*Journal of Computational Physics*, 448 (2022), paper No. 110699.)